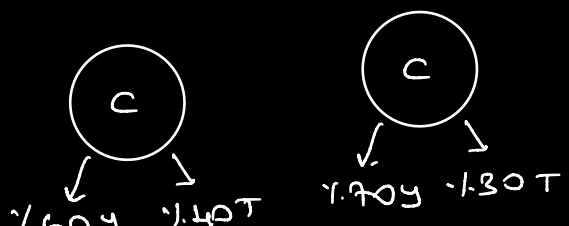
2 hileli poco



Kural

1. poco T durumunda 2 poco atilcak

	Y H	T T
Y H	0.6	0.7
T T	0.4	0.3

$$O = \begin{bmatrix} 0.6 & 0.7 \\ 0.4 & 0.3 \end{bmatrix}$$

Örn: Bu atim bu kural dahilinde yapilirsa, YIT olasiligi?

$$O \cdot V_y = V_1$$

$$\begin{bmatrix} 0.6 & 0.7 \\ 0.4 & 0.3 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

$$O \cdot V_T = V_1$$

$$\begin{bmatrix} 0.6 & 0.7 \\ 0.4 & 0.3 \end{bmatrix}\cdot\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$$

* bu durum iki kere tekrarlanirsa

$$\begin{bmatrix} 0.6 & 0.7 \\ 0.4 & 0.3 \end{bmatrix}\cdot\begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.6\times0.6+0.7\times0.4 \\ 0.4\times0.6+0.3\times0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.36+0.28 \\ 0.24+0.12 \end{bmatrix} = \begin{bmatrix} 0.64 \\ 0.36 \end{bmatrix}$$

* Bir kuantum bilgisayar her sekilde olma olasiligini hesaplar

$$|P_1 P_2\rangle = \begin{bmatrix} P_1P_2 \\ P_1 \end{bmatrix} \rightarrow$$ Tek seferde ayni ondo hesaplar

8. hafta Ders Notu

Örnek: $B = \begin{pmatrix} 0.4 & 0.6 & 0 \\ 0.2 & 0.1 & 0.7 \\ 0.4 & 0.3 & 0.3 \end{pmatrix}$ (Tasinan) olasilik matrisi verilmis olsun

* Birinci durumdan ikinci duruma gecis satiri = 0.4, 0.6, 0
* 3 → 1 gecme [0.4, 0.3, 0.3]

$$B = \begin{pmatrix} 0.4 & 0.6 & 0 \\ 0.2 & 0.1 & 0.7 \\ 0.4 & 0.3 & 0.3 \end{pmatrix}\begin{bmatrix} 0.1 \\ 0.3 \\ 0.6 \end{bmatrix} = ?$$

Iki olasilikasal Bit

$P_{b_1} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$ → Tek olasilikasal bit

$P_{b_2} = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$ → 0, → 1

$P_{b_2} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$ → 0, → 1 Tek bir sistemde birlestirirsek

$P_{b_1}P_{b_2}$		
0 0 → 00	olma	olasiligi
0 1 → 01	olma	olasiligi
1 0 → 10	"	"
1 1 → 11	"	"

Tensor carpimi ile

$$P_{b_1}P_{b_2} = P_{b_1} \otimes P_{b_2}$$

$$= \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} \otimes \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

$$|P_{b_1}P_{b_2}\rangle = \begin{bmatrix} 0.12 \\ 0.08 \\ 0.48 \\ 0.32 \end{bmatrix}$$

- Bu birlesik sistemde 00 olma ihtimali %12'dir.
- Bu birlesik sistemde 01 olma ihtimali %8 dir.
- Bu birlesik sistemde 10 olma ihtimali %48 dir.
- Bu birlesik sistemde 11 olma ihtimali %32 dir.

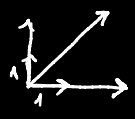
Kompozite Sistemler

$$P_{b_1} = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} = 0.2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0.8 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- bir vektör birim vektörler türünde ifade edilebilir.

$$\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}$$

birim vektör



$$P_{b_1} = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} = 0.2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0.8 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

birim(bos) vektör

$$= 0.2\,|0\rangle + 0.8\,|1\rangle$$ → kuantum diliyle tanımlar

$$\begin{bmatrix} 0.12 \\ 0.08 \\ 0.48 \\ 0.32 \end{bmatrix} = P_{b_3} = 0.12\,|00\rangle + 0.08\,|01\rangle + 0.48\,|10\rangle + 0.32\,|11\rangle$$

$$= 0.12 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 0.08 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + 0.48 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + 0.32 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Örnek: $|00\rangle = |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Sadeleştirmek amacıyla

$$P_{b_1} = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} \quad P_{b_2} = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$$

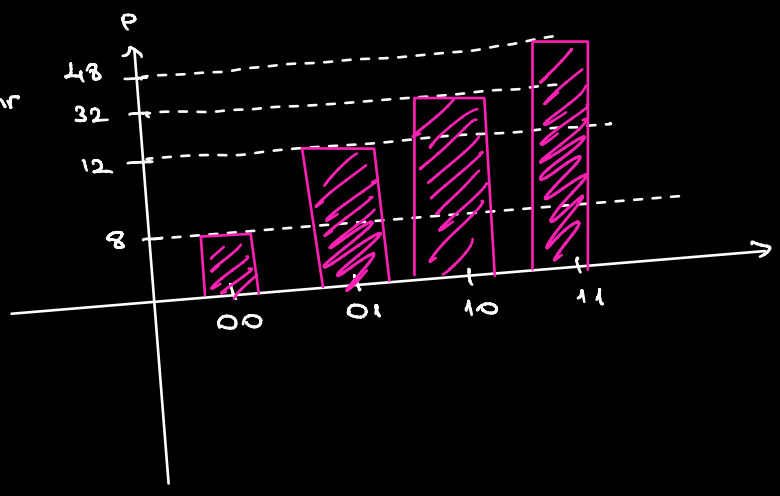
$$[0.2|0\rangle + 0.8|1\rangle] \otimes [0.6|0\rangle + 0.4|1\rangle$$

$$= [0.2 \times 0.4\,|0\rangle \otimes |0\rangle] + [0.6 \times 0.2\,|0\rangle \otimes |1\rangle + [0.8 \times 0.4\,|1\rangle \otimes |0\rangle] + [0.8 \times 0.6\,|1\rangle \otimes |1\rangle]$$

$$= [0.08\,|00\rangle + 0.12\,|01\rangle + 0.32\,|10\rangle + 0.48\,|11\rangle$$

Kural

- Sayılar sayılarla çarpılır
- Ketler ketlerle



TS [a,b,c] durum vektörünü tanımla

$|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$

$|110\rangle$ ketinin vektörel karşılığı nedir.

$|1\rangle \otimes |1\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

↳ Kuantumda dolanıklığı ifade eder.

Sword Kuantumdaki soruları çözümünü not ekle

$$M \cdot \begin{bmatrix} 0,3 \\ 0,7 \end{bmatrix} = \begin{bmatrix} 0,52 \\ 0,42 \end{bmatrix}$$

$$\underset{A}{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} \cdot \underset{b}{\begin{bmatrix} 0,3 \\ 0,7 \end{bmatrix}} = \underset{y}{\begin{bmatrix} 0,52 \\ 0,42 \end{bmatrix}}$$

$a \times 0,3 + 0,7b = 0,52$

$c \times 0,3 + d \times 0,7 = 0,42$